

Can diffusion fill in what a few photos leave out?

A study of training Gaussian Splatting from a handful of views and using Stable-Diffusion to synthesize the rest — and the two experimental mistakes that had to be undone before the real answer showed up.

object: scale-model Abrams

splatfacto / gsplat

SD-1.5 inpaint · SVD · Zero123

RTX 4060 · 8 GB

object-masked PSNR

01 The question

Gaussian Splatting is spectacular with hundreds of views and falls apart with a handful. The premise here: if we only have **5, 10, or 20** real photos of an object, can we **generate** the missing supervision with a diffusion model and recover the quality we'd get from a full capture?

The experiment is a grid. Few-shot size $N \in \{5, 10, 20\}$ crossed with a **synthetic-to-real ratio** $\{0, 25, 50, 100, 200\}\%$, crossed with an augmentation **strategy**. A full run on all views is the ceiling. Every score is **object-masked PSNR** — we measure the tank, never the background — so a model gets no credit for an easy backdrop.

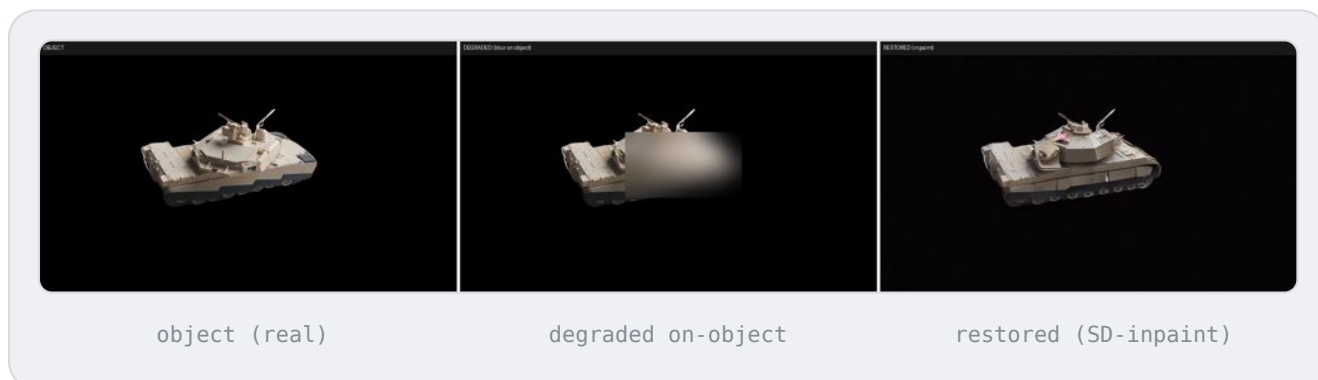
The headline result is not the one we expected. Getting to it meant discovering that *two* of our own setup choices were quietly deciding the outcome.

02 How we used diffusion

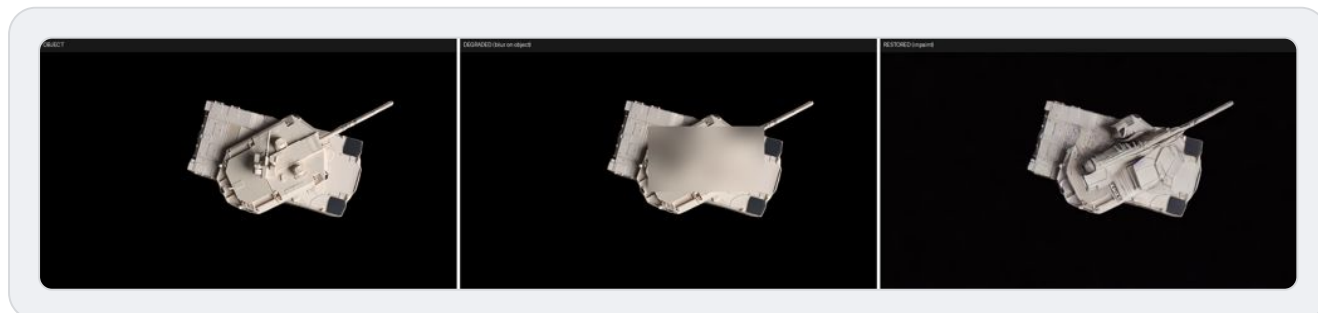
Two families of synthetic view, plus one that failed instructively. The guiding principle, learned early: **manipulate-then-restore** beats hallucinate. The most useful synthetic frame is one anchored to something real.

Inpaint — restore degraded detail at a real pose

The workhorse. Take a real training view, **degrade a region that sits on the object** (blur or cut it, confined to the object's mask so we never touch background), then ask SD-inpainting to **restore** it. The result is a frame at the exact real camera pose, photometrically consistent, that adds a second, sharpened look at detail the optimizer would otherwise see only once. It regularizes; it doesn't invent viewpoints.

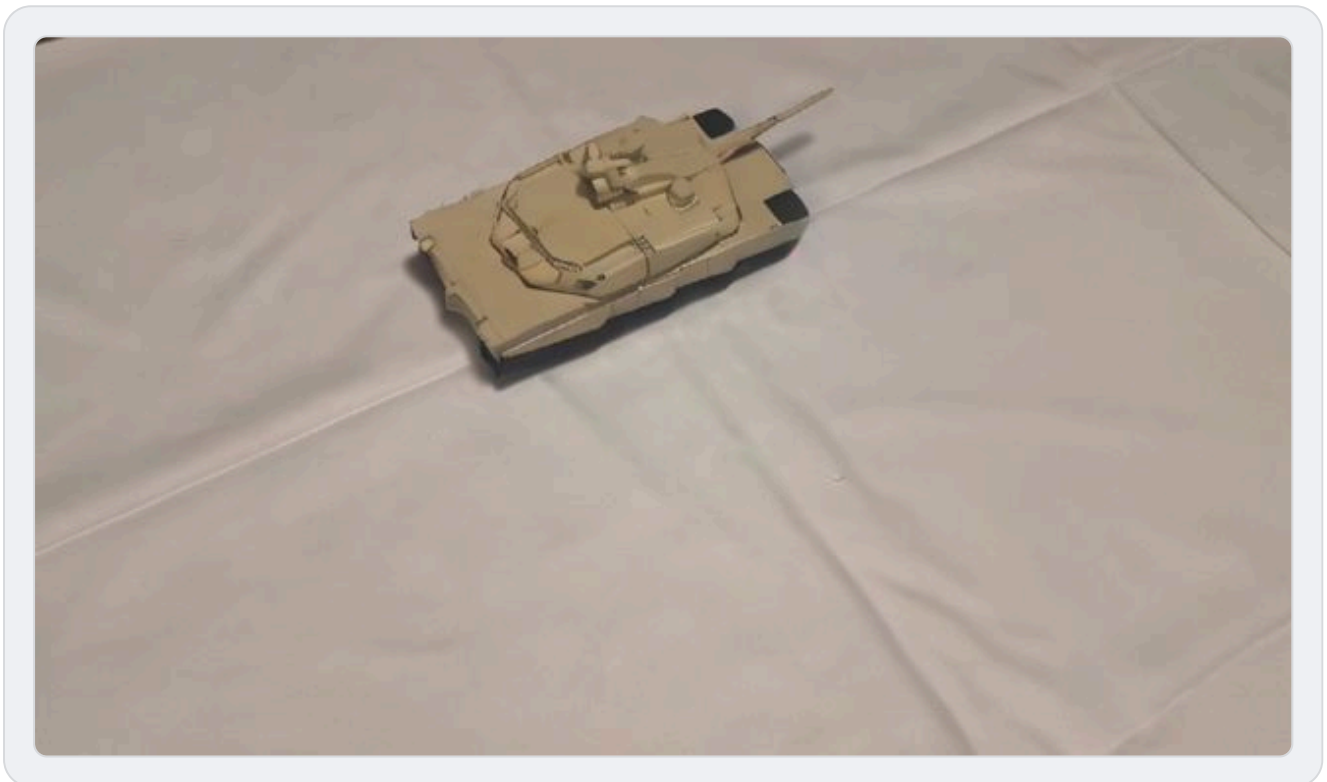


The inpaint process. A patch on the turret is blurred, then diffusion restores plausible sharp detail – masked to the object so the background is untouched. This runs at every few-shot pose to build the synthetic pool.



SVD — genuine novel viewpoints, with poses we can trust

Inpaint never adds *coverage*. For that we extended the capture with **Stable Video Diffusion**: seed a short clip from a real view to get nearby novel angles. The hard part is the camera pose of a generated frame — our earlier attempt (**Zero123**) hand-computed it and the reconstruction diverged. The fix was to stop computing poses: run **COLMAP on the real + generated frames together** and let it solve every pose jointly. 271/280 generated frames registered onto the orbit; the recovered intrinsics matched the real camera to ~1%.



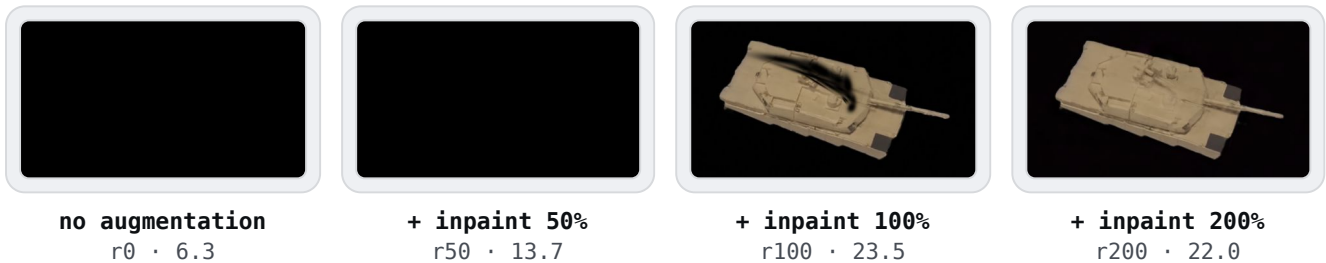
An SVD novel view at a COLMAP-assigned pose. Convincing in isolation — which, as §4 shows, turns out to be exactly the trap.

03 The story — two mistakes we made

We built the scene **object-centric**: segment the tank onto a black background before COLMAP, so the point cloud is object-only and the metric is clean. On that setup, inpaint looked *fantastic*.

Segmented grid — the first, misleading result · full 25.0									
N	base	i25	i50	i100	i200	s25	s50	s100	s200
n5	6.7	7.7	9.5	13.9	17.5	.	.	.	.
n10	6.9	8.6	10.0	18.6	19.6	.	.	7.0	6.5
n20	6.3	9.4	13.7	23.5	22.0	.	.	6.3	6.2

Read the inpaint columns: n5 climbs 6.7 → 17.5, n20 6.3 → 23.5 — a large, monotonic, visually obvious gain, right up near the full = 25.0 ceiling. We nearly wrote the paper here. Meanwhile **Zero123 diverged** (~6, below baseline) and **SVD came out neutral** even with correct poses. The tidy story: *same-pose diffusion rescues few-shot, novel-view diffusion doesn't.*



Same held-out view, **segmented n20**, as the inpaint ratio rises — the object literally climbs out of the black. Dramatic, and exactly why we nearly stopped here. (r0/r50 brightened for visibility.)

Then one control test broke it.

MISTAKE 1 · THE SEGMENTATION

Trained on the *same* 20 views but with the natural background left in, the baseline jumped from 6.34 to 27.4 — without any augmentation at all.



Object-on-black, 20 views (brightened 3× to show it). The model renders almost nothing — a foggy smear. PSNR 6.3.



Natural background, same 20 views. A clean tank — PSNR 20.9.

Gaussian Splatting leans on **background texture** to anchor a sparse optimization — the fabric gives the cameras and Gaussians something to lock onto. Segmenting to black removed it and **destabilized few-shot**. So the celebrated inpaint gains were largely *recovery from a handicap we had inflicted*: dense supervision re-stabilizing a broken baseline. We archived the segmented pipeline and re-ran everything on natural backgrounds — training on the full image, still scoring only the object.

MISTAKE 2 · THE POINT-CLOUD INIT

The init cloud came from COLMAP on all 320 views — so every "few-shot" run was secretly handed the complete object geometry. Worth about +7 PSNR.

The ply leak — random vs full-320 ply baseline

N	random	+ full ply	leak
n5	14.39	20.95	+6.56
n10	14.77	24.33	+9.56
n20	20.91	28.10	+7.19

Strip the leaked prior (random init) and a n5 baseline drops 20.95 → 14.39. A clean calibration falls out: **20 honest views ≈ 5 views plus the full-scene ply**. Only with *both* confounds removed — natural background, random init — is the question finally honest.

04 The honest answer

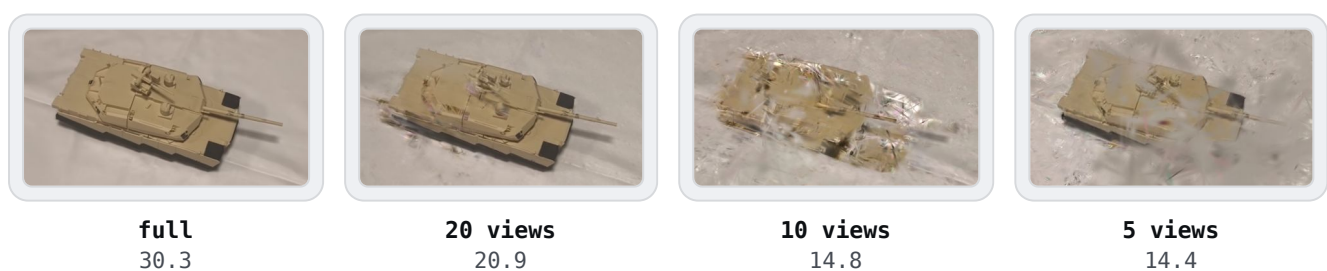
Below: the dose-response across every cell, on the same y-axis. Left column is the old segmented setup (inpaint rocketing off a broken ~ 6 baseline); right column is the honest natural-bg setup (baselines already at 14–21, far less to add).

First, the renders themselves — the same held-out view, reconstructed by each model (natural background). You can read the quality order straight off the images:



Inpaint sharpens the baseline; **SVD softens** it. The dilution is visible — the SVD render is the blurriest of the reconstructions, exactly as its PSNR says.

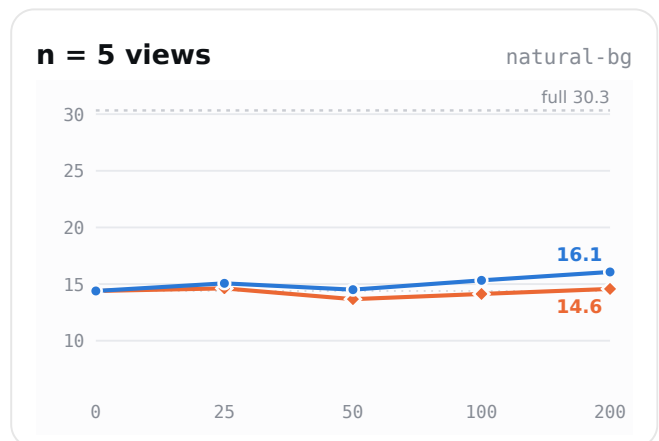
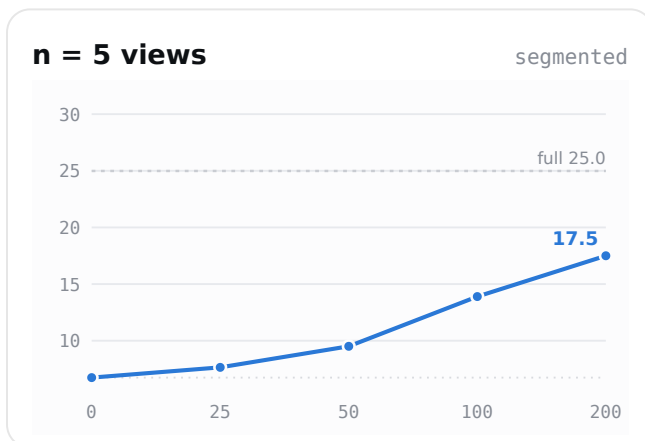
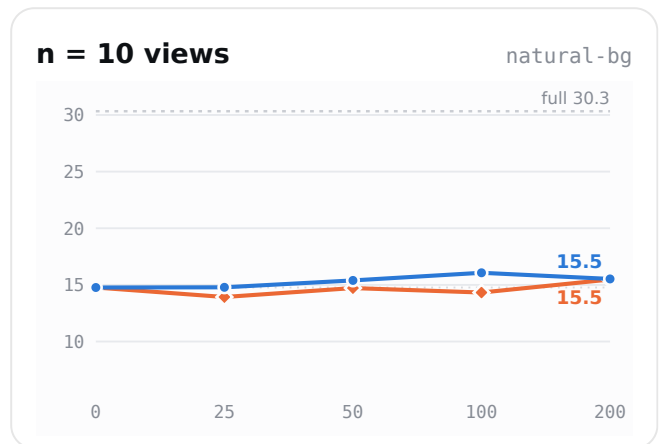
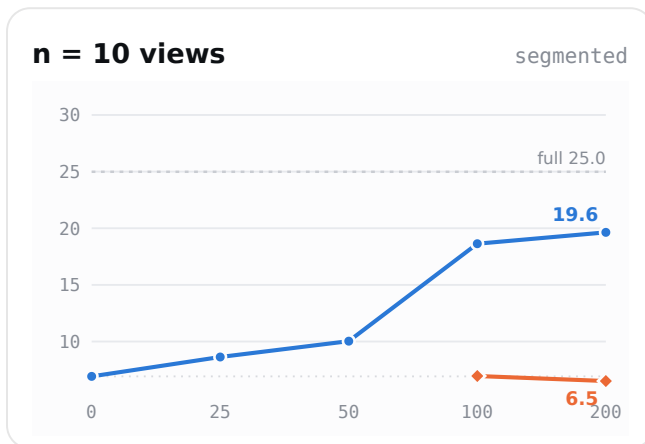
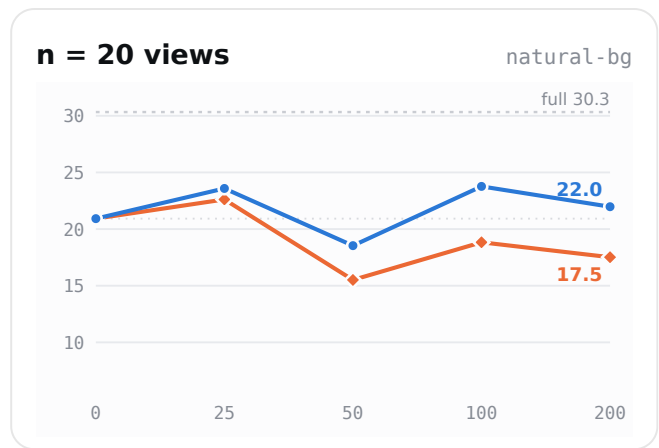
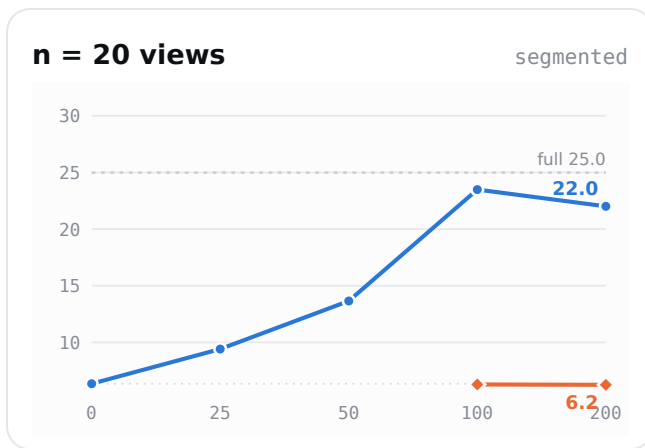
And how the bare reconstruction degrades as real views get scarcer — the gap augmentation is trying to close:



Honest baselines (random init), no augmentation. Fewer real views → rougher tank.

Quantified across every cell, on a shared y-axis:

— inpaint — SVD --- baseline (r0) & full ceiling (dashed)



Honest natural-bg grid (random init) · full 30.3

N	base	i25	i50	i100	i200	s25	s50	s100	s200
n5	14.4	15.1	14.5	15.3	16.1	14.6	13.7	14.1	14.6
n10	14.8	14.8	15.4	16.1	15.5	13.9	14.7	14.3	15.5
n20	20.9	23.6	18.5	23.8	22.0	22.6	15.5	18.8	17.5

Inpaint still helps, everywhere, peaking around a 1:1 ratio (n20 +2.9).
SVD is dose-dependent: a small amount helps at n20 (+1.7) but larger amounts **dilute and hurt** (−2.1, −3.4). And it's *not a bug* — poses, intrinsics

and resolution were all verified. SVD frames are simply **soft diffusion output** next to sharp real photos, so piling them on drags the reconstruction toward that softer average.

THE UNIFYING INSIGHT

Augmentation's value inverts with init quality — synthetic frames help a poor init (they add missing supervision) and dilute a good one (the geometry is already there).

Augmentation Δ vs init quality (n20)				
n20	poor init (random)	Δ	good init (own-ply)	Δ
baseline	20.91		26.86	
+ inpaint r100	23.76	+2.85	25.32	-1.54
+ SVD r100	18.83	-2.08	20.60	-6.26

With random init (no geometry) inpaint adds real signal and helps. Give the model a good init — the 20-view own-ply, which nearly matches the full ply — and the same frames only soften it. This is the whole arc in one sentence: *inpaint "worked" earlier because the baseline was broken; fix the baseline and there is little left to add.*

05 What it means

The reconstruction the corrected pipeline actually produces, for reference:



Held-out novel view, natural-bg reconstruction (PSNR 30.3 ceiling).
Sharp hull, turret and barrel – nothing like the segmented collapse.

- **Diffusion augmentation does help few-shot object GS** — but as clean *same-pose regularization* (inpaint), not as novel-view synthesis.
- **Novel-view synthesis (Zero123, SVD) does not help** and can hurt — even with geometrically-correct poses — because generated views are softer and not mutually 3D-consistent enough to reinforce a reconstruction.
- **Setup dominates.** Object-on-black segmentation and a leaked full-scene ply each moved the numbers more than any augmentation did. The real lesson of the project is methodological: measure the confounds before crediting the method.
- **Augmentation is worth most exactly where geometry is worst** — poor init, few real views. Where a good prior exists, leave the synthetic frames out.

Diffusion can fill in what a few photos leave out — as long as what it fills in is a *sharper look at what was already there*, not a guess at what wasn't.

Pipeline · segment/extract → COLMAP → assemble (real + synthetic) → splatfacto @30k → masked eval → registry

Diffusion · SD-1.5 inpainting (object-masked degrade→restore) · Stable Video Diffusion (img2vid, COLMAP-solved poses) · Zero123-xl (deprecated arm)

Metrics · object-masked PSNR/SSIM/LPIPS · held-out real views · pose_fit_residual ≈ 0

Hardware · RTX 4060 8 GB · gsplat +pt21cu118 · diffusers 0.27.2